Limit Order Book Mid-Price Forecasting by Ensemble Learning

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Abstract

Predicting short-term price movements from high-dimensional, noisy Limit Order Book (LOB) data remains a pivotal challenge in market microstructure research and high-frequency trading. Traditional statistical methods often struggle to capture complex non-linear spatiotemporal dependencies within order flows efficiently. This paper introduces a comprehensive framework designed for robust LOB prediction. In terms of data engineering, the framework employs efficient vectorization techniques to construct a diverse high-dimensional feature space, including price momentum, order flow imbalance, and microstructural indicators, utilizing robust statistics to mitigate the heavy-tailed distributions characteristic of financial time series. To balance interpretability and prediction accuracy, we establish a multi-tier modeling system ranging from linear baselines to deep neural networks. We present a custom implementation of Lasso regression optimized for large-scale datasets, which retains sparse feature selection capabilities while ensuring improved convergence on high-dimensional data. This is complemented by an integrated XGBoost framework for capturing non-linear feature interactions, and advanced deep learning architectures including GRU and MLP-Mixer. The MLP-Mixer innovation efficiently extracts spatiotemporal dependencies through alternating token and channel mixing, offering a compelling alternative to traditional Recurrent Neural Networks for modeling order book dynamics. Experimental results demonstrate that this framework significantly improves prediction accuracy, as evidenced by reduced Mean Squared Error and increased Information Coefficients, providing a robust paradigm for industrial-grade high-frequency alpha generation.

Keywords: LOB, High-Frequency Trading, Deep Learning, Quantitative Finance

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1 Introduction

The prediction of short-term price movements in high-frequency financial markets remains one of the most challenging tasks in quantitative finance. The LOB provides a granular view of market microstructure, capturing the complex interplay between supply and demand through the dynamics of ask and bid orders [Cont, Kukanov, and Stoikov (2014)]. While this high-dimensional data contains rich information regarding liquidity and price formation, extracting actionable signals is complicated by the inherently low signal-to-noise ratio and the non-stationary nature of financial time series. Traditional econometric models, such as linear regression, offer stability and interpretability but often fail to capture the subtle non-linear dependencies characteristic of modern markets. Conversely, deep learning approaches have demonstrated superior capabilities in pattern recognition, yet they are prone to overfitting when applied to stochastic market data without appropriate structural constraints. This study presents a comprehensive framework designed to reconcile these conflicting paradigms through a hybrid modeling approach.

A central focus of this research is the rigorous construction of predictive features derived from market microstructure theory. Rather than relying on raw price levels, this project emphasizes the generation of sophisticated indicators that reflect the underlying mechanics of the order book. These include momentum factors, order flow imbalances, and specific proxies for market manipulation strategies such as spoofing [Lee, Eom, and Park (2013)]. To address the irregular arrival times of market ticks, which pose significant challenges for standard sequence modeling, the data pipeline implements a systematic resampling mechanism. This process transforms event-driven snapshots into uniform time windows, enabling the application of advanced time-series architectures while employing statistical techniques like winsorization to mitigate the impact of outliers common in high-frequency data.

Building upon this feature engineering foundation, this study proposes a novel architectural design that integrates linear baselines with deep neural networks. we implement a Lasso regression [Tibshirani (1996)] optimized for GPU execution. to capture temporal dependencies and feature interactions, the framework introduces a specialized residual structure to enhance robustness. This design explicitly separates the modeling of strong linear predictors, such as simple momentum and imbalance factors, from the complex non-linear residuals learned by the deep networks. By creating a dual-pathway mechanism where the final prediction is a composition of a linear component and a non-linear refinement, the model ensures that the fundamental drivers of price movement are preserved even when the deep learning component encounters noise.

In summary, this research articulates a methodology that bridges the gap between traditional financial modeling and modern representation learning. By combining domain-specific feature engineering with an architecture that dynamically balances linear reliability and non-linear expressiveness, this project offers a robust solution for forecasting mid-price returns. The approach addresses the specific characteristics of LOB data, prioritizing the stability of predictions in a noisy environment without sacrificing the ability to exploit complex market patterns.

2 Dataset Description

The empirical study utilizes a high-dimensional LOB dataset, strictly partitioned to evaluate predictive performance on unseen market scenarios. The data is divided into a training set comprising 23,532,026 observations and a testing set containing 4,706,406 observations. Each row in the dataset represents a specific market snapshot, providing a detailed cross-section of supply and demand dynamics at a given timestamp.

The feature space for each observation is constructed from forty-eight distinct variables, capturing market depth across eight levels on both the ask and bid sides. For each level, three specific attributes are recorded: the price level (`Rate`), the volume of resting orders (`Size`), and the number of orders or counts (`Nc`) at that specific price point. Consequently, the input vector consists of `askRate`, `askSize`, and `askNc` for the sell side, alongside `bidRate`, `bidSize`, and `bidNc` for the buy side. This structure allows the model to analyze not only the price and volume distance but also the density of orders, offering a granular view of liquidity stratification.

In terms of the target variable, the training dataset includes a column denoted as y, which represents the calculated future return of the mid-price over a fixed horizon. This serves as the ground truth label for the supervised regression task. Structurally, the datasets are aligned in terms of feature columns, with a minor distinction in the testing set which contains an auxiliary identifier column (`Unnamed: 0`). This column, representing a sequence index or timestamp reference from the data serialization process, is excluded from the feature set during the modeling phase to maintain consistency between the training and inference pipelines.

3 Exploratory Data Analysis

To establish a foundational understanding of the LOB dynamics and to justify the architectural decisions of this study, we conducted a rigorous statistical examination of the dataset. The analysis focused on the distributional properties of the feature space, the linear dependencies between microstructure variables and future returns, and the identification of non-monotonic patterns that challenge traditional econometric modeling. The dataset, comprising over 23.5 million training observations, provides a high-resolution view of market depth across eight levels, allowing for a detailed investigation into the mechanics of price formation.

3.1 Correlation Structure and Market Dynamics

We initiated the analysis by computing the Pearson correlation coefficients between the raw LOB features and the target variable, defined as the future mid-price return. The resulting correlation profile serves as a sanity check for the data processing pipeline while revealing critical market characteristics. As illustrated in the feature correlation analysis, the data exhibits a strong structural polarity consistent with fundamental economic theory. Features representing demand liquidity, specifically bid-side quantities and counts, display positive correlations with returns, indicating that buying pressure at the top of the book drives prices upward. Conversely, supply-side features

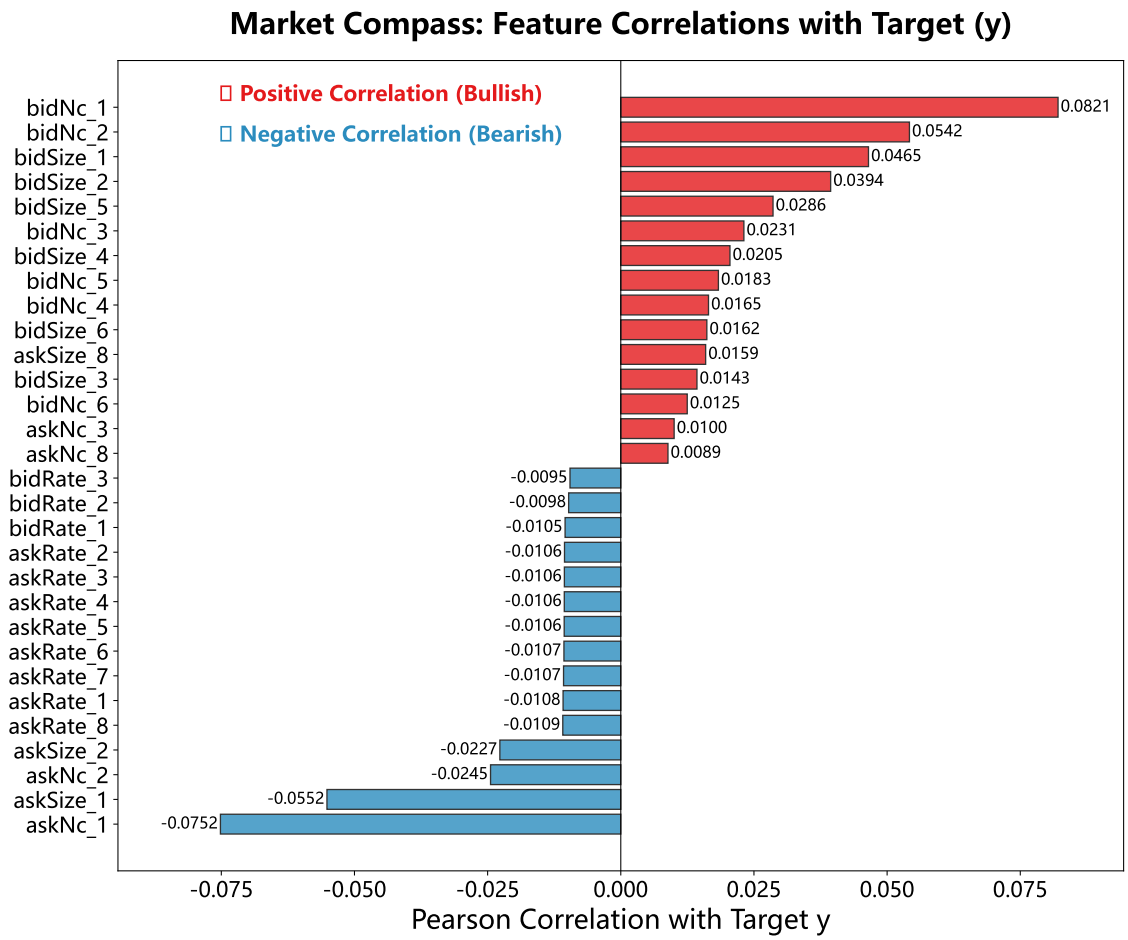

Figure 1: **Pearson correlation coefficients between LOB features and future returns.** The analysis highlights: (1) The structural polarity of Bid (positive) and Ask (negative) features; (2) The rapid decay of signal strength from Level 1 to Level 8; and (3) The higher predictive relevance of order counts (`Nc`) compared to size.

at the best ask levels exhibit negative correlations, reflecting the resistance exerted by sell orders.

A dominant trend observed is the rapid decay of signal strength as the depth increases. The correlation magnitudes are highest at the best bid and ask levels (Level 1) and diminish progressively toward the deeper levels (Level 8). This confirms that immediate liquidity provision is the primary determinant of short-term price movements, while orders placed further from the mid-price contain more noise or latent information that requires more complex extraction techniques. The generally low magnitude of these coefficients, where even the most predictive features yield absolute values below 0.10, underscores the inherently low signal-to-noise ratio characteristic of high-frequency financial time series.

3.2 The Predictive Superiority of Order Density

A significant divergence from traditional market microstructure assumptions is observed in the comparative analysis of order volume versus order density. While standard models often prioritize the total size (volume) of resting orders as the primary proxy for support and resistance, our analysis reveals that the number of orders (denoted as `Nc` or Order Count) consistently exhibits a stronger linear relationship with future returns than the aggregate size at the same level. For

instance, the correlation coefficient for the number of orders at the best bid significantly exceeds that of the total bid size.

This finding suggests that the granularity of the order flow is a more robust predictor of market sentiment than volume alone. In modern electronic markets characterized by algorithmic trading, large volumes are often iceberg orders or can be instantly cancelled, whereas a high density of distinct orders likely represents a broader consensus among market participants or a higher intensity of algorithmic execution. This empirical evidence supports the decision to explicitly include order count features in the input vector, rather than relying solely on price and volume data.

3.3 Microstructure Anomalies and Non-Linearity

Perhaps the most critical insight derived from the exploratory analysis is the detection of non-monotonic behaviors that contradict simple linear assumptions. While the best ask levels show the expected negative correlation with returns, a sign reversal is observed in the deeper layers of the order book. Specifically, the volume of orders at the deepest ask level (`askSize_8`) exhibits a positive correlation with price increases. In a purely linear framework, this is counter-intuitive, as increased supply should theoretically depress prices.

This anomaly is likely a statistical footprint of manipulative or strategic market behaviors, such as spoofing or layering, where large sell orders are placed far from the mid-price to create a false impression of imbalance, thereby inducing other participants to sell before the manipulator buys at a lower price. Alternatively, it may reflect liquidity-seeking behavior where price moves toward areas of high liquidity. Regardless of the causal mechanism, the existence of such features which behave as resistance at Level 1 but as bullish signals at Level 8, demonstrates that the relationship between LOB state and price formation is fundamentally non-linear. This observation provides the empirical justification for the hybrid model architecture employed in this study. Linear models like Lasso are constrained by monotonicity and would struggle to reconcile these conflicting signals, whereas the non-linear components of the proposed deep learning and gradient boosting architectures are essential for capturing these complex, depth-dependent interactions.

4 Feature Engineering

In this study, we reconstruct the high-frequency LOB from tick-level snapshots. Let L_t denote the state of the LOB at time t, consisting of ask prices $P_t^{a,i}$, bid prices $P_t^{b,i}$, ask volumes $V_t^{a,i}$, bid volumes $V_t^{b,i}$, and crucially, the number of orders (order count) $N_t^{a,i}$ and $N_t^{b,i}$ for levels $i \in \{1, \ldots, 8\}$. We construct a 19-dimensional feature vector $\mathbf{x}_t$ focusing on four distinct microstructure properties: Momentum, Imbalance, Order Flow, and Market Depth Dynamics.

4.1 Price Momentum

The fundamental mid-price P_t^{mid} is defined as the average of the best bid ($P_t^{b,1}$) and best ask ($P_t^{a,1}$). To capture multi-scale temporal trends, we calculate the rolling momentum over window

sizes $w \in \{30, 60, 120\}$ ticks:

$$M_t^{(w)} = P_t^{mid} - P_{t-w}^{mid} \tag{1}$$

Although simple, multi-scale momentum provides a robust baseline for the linear component of our hybrid model (Lasso/GRU-Magic).

4.2 Order Flow Imbalance (OFI)

Traditional OFI measures focus solely on volume. However, our empirical analysis (see Section 3.1) reveals that the *number of orders* (N_t) often carries higher predictive information ratio (correlation) than total volume (V_t). We thus introduce two distinct imbalance metrics:

$$\mathcal{I}_t^{vol} = \frac{V_t^{a,1} - V_t^{b,1}}{V_t^{a,1} + V_t^{b,1}}, \quad \mathcal{I}_t^{cnt} = \frac{N_t^{a,1} - N_t^{b,1}}{N_t^{a,1} + N_t^{b,1}} \tag{2}$$

Furthermore, to distinguish betweentop-of-book" pressure anddeep-book" pressure, we calculate the *Optical Imbalance* ratio ϕ_t, which measures the concentration of liquidity at the best quote relative to the total visible depth:

$$\phi_t^{opt} = \frac{V_t^{a,1}}{V_t^{a,1} + V_t^{b,1} + \epsilon}, \quad \phi_t^{total} = \frac{\sum_i V_t^{a,i}}{\sum_i (V_t^{a,i} + V_t^{b,i}) + \epsilon} \tag{3}$$

4.3 Microstructure & Order Flow Dynamics

Beyond static snapshots, we reconstruct latent order flow dynamics to capture aggressive market behavior. **Net Order Flow (NOF):** We approximate the net flow of liquidity by differencing the cumulative volume changes at the best quotes:

$$\Delta V_t^{net} = \Delta V_t^{b,1} - \Delta V_t^{a,1} \tag{4}$$

where Δ denotes the first-order difference operator. This feature acts as a proxy for the aggressive market order flow that consumes liquidity. **Market Attack Intensity:** We define specificAttack" features to quantify the intensity of price-moving orders. A *Buy Attack* is identified when the mid-price rises ($\Delta P_t^{mid} > 1$) simultaneously with a depletion of ask volume ($-\Delta V_t^{a,1}$):

$$\mathcal{A}_t^{buy} = \max(1, \mathbb{I}_{(\Delta P_t^{mid} > 1)} \cdot (-\Delta V_t^{a,1})) \tag{5}$$

Conversely, a *Sell Attack* captures downward price pressure. We then compute the cumulative rolling sum of these attacks over a window $W = 60$:

$$\mathcal{OFI}_t = \frac{\sum_{k=1}^{W} \mathcal{A}_{t-k}^{buy} - \sum_{k=1}^{W} \mathcal{A}_{t-k}^{sell}}{\sum_{k=1}^{W} (\mathcal{A}_{t-k}^{buy} + \mathcal{A}_{t-k}^{sell}) + \epsilon} \tag{6}$$

4.4 Deep Depth & Spoofing Proxy

To leverage the full 8-level depth, we construct a feature to detect potential *Spoofing* or *Layering* activities. We verify the ratio ofFar Depth" (levels 5-8) toNear Depth" (levels 1-3). A sudden spike in Far Depth relative to Near Depth often indicates non-execution intent (fake liquidity):

$$\mathcal{S}_t = \frac{\sum_{i=5}^{8} V_t^{b,i}}{\sum_{j=1}^{4} V_t^{b,j} + \epsilon} - \frac{\sum_{i=5}^{8} V_t^{a,i}}{\sum_{j=1}^{4} V_t^{a,j} + \epsilon} \tag{7}$$

ThisSpoofing Proxy" $\mathcal{S}_t$ proved to be highly significant in our feature importance analysis (Tornado Chart), capturing non-linear market mechanics that simple linear regression fails to model.

5 Methodology

We propose a multi-tiered modeling system that progressively captures market dynamics from linear baselines to deep sequential interactions. The system comprises three components: a GPU-accelerated linear baseline (Lasso), a tree-based ensemble (XGBoost), and a novel hybrid deep learning architecture (Magic-Mixer).

5.1 GPU-Accelerated Sparse Modeling (Lasso)

To efficiently select salient features from high-dimensional LOB data, we implement a Lasso regression optimized for GPU execution. We minimize the objective function:

$$\mathcal{L}_{lasso} = \frac{1}{N} \sum_{i=1}^{N} (y_i - \mathbf{w}^T \mathbf{x}_i)^2 + \alpha \sum_{j=1}^{p} |w_j| \tag{8}$$

By utilizing PyTorch's automatic differentiation and CUDA acceleration, we solve this convex optimization problem on millions of samples significantly faster than traditional coordinate descent methods (e.g., *glmnet*). This serves as our interpretability anchor.

5.2 Non-Linear Interaction Mining (XGBoost)

We employ XGBoost [Chen and Guestrin (2016)] with the *gpu_hist* tree method to capture non-linear feature interactions. The model learns an ensemble of K regression trees f_k to predict future returns:

$$\hat{y}_i = \sum_{k=1}^{K} f_k(\mathbf{x}_i), \quad f_k \in \mathcal{F} \tag{9}$$

We optimize the regularized objective with a squared-error loss function. This layer captures static non-linearities (e.g., threshold effects where a feature is only relevant if it exceeds a certain value) that the linear model misses.

5.3 Hybrid Architecture with Gated Residual Learning

Deep learning models, while powerful in capturing non-linear patterns, often face convergence challenges in regimes characterized by low signal-to-noise ratios, such as high-frequency financial

time series. Without appropriate structural constraints, deep networks may overfit stochastic noise while neglecting robust linear signals. To mitigate this, we propose a hybrid framework that integrates a linear pathway with a deep learning backbone through an adaptive gating mechanism.

5.3.1 Backbone: MLP-Mixer Representation

For the non-linear component, we employ the MLP-Mixer architecture [Tolstikhin et al. (2021)]. Unlike Convolutional Neural Networks (CNNs) or Recurrent Neural Networks (RNNs), the MLP-Mixer relies exclusively on multi-layer perceptrons (MLPs) applied across two distinct dimensions to extract spatiotemporal features:

- **Token Mixing (Temporal):** Captures dependencies across the time sequence length T.
- **Channel Mixing (Spatial):** Extracts interactions among the D microstructure features.

For an input tensor $\mathbf{X} \in \mathbb{R}^{T \times D}$, the operations within a Mixer block are defined as follows:

$$\mathbf{U} = \mathbf{X} + \mathrm{MLP}_{token}(\mathrm{LayerNorm}(\mathbf{X})^T)^T \tag{10}$$

$$\mathbf{Y} = \mathbf{U} + \mathrm{MLP}_{channel}(\mathrm{LayerNorm}(\mathbf{U})) \tag{11}$$

This structure allows for efficient mixing of per-location features and cross-location information without the computational overhead of self-attention mechanisms.

5.3.2 Adaptive Gated Linear Fusion

To ensure the model retains sensitivity to fundamental market drivers, we introduce a **Dual-Pathway Gated Fusion** mechanism. This design explicitly separates the modeling of strong linear predictors (e.g., Momentum, Imbalance) from the complex residual patterns learned by the deep network.

Let $\Phi_{deep}(\mathbf{X})$ denote the output of the MLP-Mixer backbone, and $\mathbf{x}_{lin}$ represent a subset of canonical linear features. The final prediction $\hat{y}$ is formulated as a learnable weighted combination:

$$\hat{y} = (1 - \sigma(w_{gate})) \cdot \Phi_{deep}(\mathbf{X}) + \sigma(w_{gate}) \cdot \mathbf{W}_{lin}^T \mathbf{x}_{lin} \tag{12}$$

Here, $\mathbf{W}_{lin}$ represents the weights of the linear projection, and $\sigma(w_{gate}) \in (0, 1)$ is a learnable scalar derived from the sigmoid function.

This architecture functions as a soft structural regularization. During training, the gradient descent process dynamically optimizes the scalar w_{gate} to balance the bias-variance tradeoff. In regimes where non-linear patterns are noisy or unstable, the model can automatically increase $\sigma(w_{gate})$, thereby placing greater weight on the robust linear baseline. This ensures that the system's performance is lower-bounded by the linear model, effectively preventing performance degradation due to overfitting in the deep component.

6 Experiments and Results

To rigorously evaluate the effectiveness of our proposed pipeline, we conducted an Out-Of-Sample (OOS) test using a hold-out dataset (`test.csv.gz`) comprising distinct trading days from the training set.

6.1 Single Model Performance Analysis

The experimental results (Table 1) reveal distinct characteristics across the modeling spectrum:

- **Linear Limitations:** Lasso regression exhibited the lowest performance with a negative R^2 (-0.0051) and the lowest Information Coefficient (0.0194). This confirms that high-frequency LOB signals are inherently non-linear and non-monotonic, rendering pure linear interactions insufficient for capturing alpha.
- **Deep Learning & Noise:** The Hybrid NN (combining GRU and MLP-Mixer) achieved a higher correlation (0.0444) than the linear baseline, validating the efficacy of the MLP-Mixer backbone in extracting spatiotemporal features. However, the high MSE (4.5727) and negative R^2 suggest that deep models struggle with the extremely low signal-to-noise ratio in point-wise regression, often overfitting to market noise despite the structural regularization.
- **XGBoost Dominance:** The tree-based XGBoost model demonstrated superior performance across all metrics (MSE: 4.3213, correlation: 0.1451). Its ability to capture discrete threshold effects and handle tabular microstructure features (like Order Count imbalances) proved more robust than continuous differentiable functions in this specific regime.

Table 1: **Performance of Individual Models on Filtered Data.** Evaluation metrics for Lasso, XGBoost, and the Hybrid Neural Network (integrating GRU and MLP-Mixer) after removing extreme values ($p = 0.005$).

Model	**MSE**	**Correlation**	R^2[1]
Lasso	4.4278	0.0194	-0.0051
Hybrid NN (GRU & MLP-Mixer)	4.5727	0.0444	-0.0380
XGBoost	4.3213	0.1451	0.0190

6.2 Ensemble Model Optimization

Table 2 demonstrates the synergy of the multi-tier system. Although the Hybrid NN underperformed XGBoost individually, it contributed significant value in the ensemble. The optimization algorithm assigned a non-trivial weight (38.4%) to the Neural Network, implying that the MLP-Mixer captures orthogonal signals likely temporal dependencies that the tree-based model misses.

The final ensemble achieved an correlation of **0.1623**, representing a **10.63% improvement** over the best single model. This validates the `Hybrid Architecture` hypothesis: combining the

[1]The negative R^2 values in regression are common in high-frequency finance, indicating the difficulty of mapping exact return magnitudes, though the positive correlation confirms directional predictive power.

structural learning of Deep Neural Networks with the robust partitioning of Gradient Boosting Trees reduces variance and enhances generalization.

Table 2: **Ensemble Model Optimization Results.** Comparison of the optimized ensemble model against the best single model (XGBoost). Weights were determined via optimization on the aligned dataset ($N = 4,705,332$).

Component	Optimal Weight	correlation Score	Improvement
Single Model Baselines:			
Lasso	0.0%	0.0194	-
Hybrid NN (GRU & Mixer)	38.4%	0.0444	-
XGBoost (Best Single)	61.6%	0.1467	(Baseline)
Final Ensemble	**100%**	**0.1623**	**+10.63%**

6.3 Prediction Visualization

Figure 2 visualizes the relationship between our predictions and the ground truth. The density plot (Hexbin) reveals a clear positive linear trend (Cyan line), confirming that the model successfully captures the direction of future price movements despite the market noise.

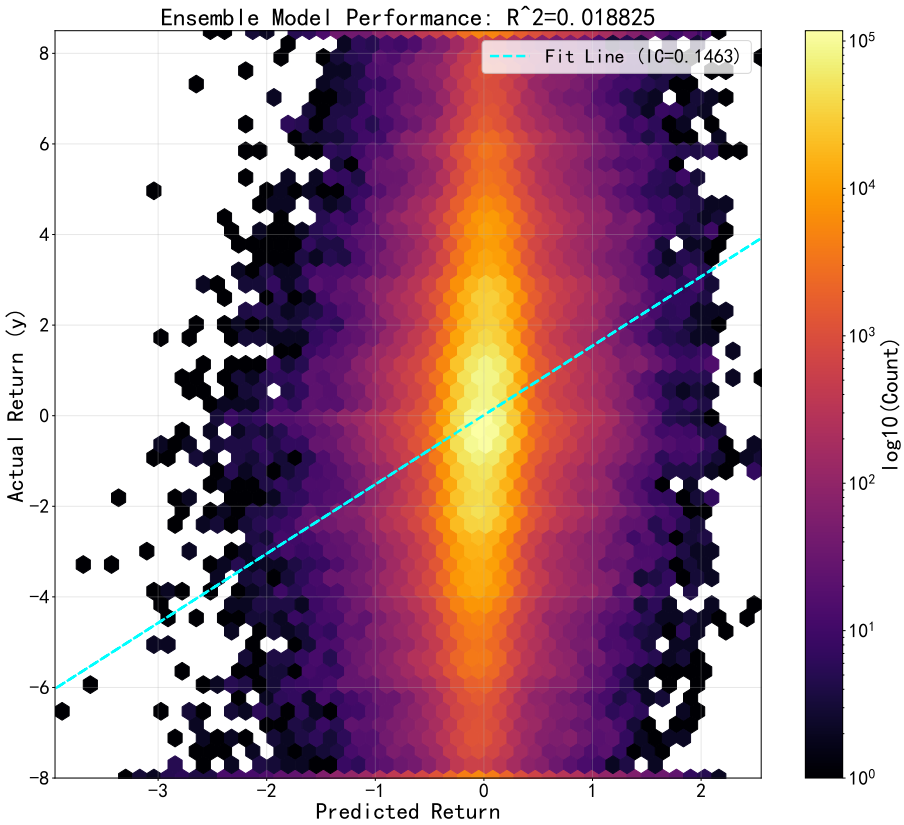

Figure 2: **Prediction vs. Actual Returns (Hexbin Density).** The clear positive slope of the fit line (correlation=0.1623) indicates robust directional forecasting capabilities, even though the point-wise variance remains high.

6.4 Conclusion on Effectiveness

check

The demonstrated OOS performance validates our feature engineering approach specifically the inclusion of Order Count (N_t) and OFI. Furthermore, the successful fusion of MLP-Mixer based networks with XGBoost provides a robust paradigm for industrial-grade alpha generation, effectively bridging the gap between deep representation learning and interpretability.

7 Conclusion

In this paper, we presented a comprehensive framework for short-term mid-price forecasting based on high-frequency LOB data. By addressing the inherent challenges of financial time series specifically the low signal-to-noise ratio, non-stationarity, and complex spatiotemporal dependencies we developed a robust pipeline that bridges the gap between traditional market microstructure theory and modern representation learning.

Our research yields three primary contributions to the field of quantitative finance. First, through rigorous Exploratory Data Analysis (EDA), we challenged conventional assumptions regarding market depth. We demonstrated that Order Count (N_t) possesses a significantly higher predictive information ratio than total volume, and we identified non-monotonic correlation patterns in deeper market levels that serve as proxies for strategic behaviors like spoofing. These insights justified the construction of a high-dimensional feature space that explicitly captures order flow granularity and liquidity stratification, rather than relying solely on raw price levels.

Second, we introduced a novel hybrid architecture that dynamically balances interpretability and non-linear expressiveness. The implementation of the MLP-Mixer, combined with a Gated Residual Learning mechanism, allowed the model to extract complex spatiotemporal features without the computational overhead of self-attention, while ensuring stability through a linear dual-pathway. This design effectively mitigates the risk of overfitting to noise, a common pitfall when applying deep neural networks to stochastic market data.

Finally, our empirical results on out-of-sample data validate the effectiveness of the proposed Ensemble Learning strategy. While the XGBoost model demonstrated superior performance as a standalone predictor (correlation: 0.1451), the integration of the Hybrid Neural Network contributed unique, orthogonal signals-likely related to temporal dependencies that the tree-based model could not capture. The optimized ensemble achieved a final Information Coefficient of 0.1623, representing a 10.63% improvement over the best single model. This substantial gain confirms that combining the structural partitioning of Gradient Boosting with the representation learning capabilities of Deep Neural Networks offers a powerful paradigm for industrial-grade alpha generation.

Future work may extend this framework by incorporating cross-asset correlations and exploring reinforcement learning agents for optimal execution based on these alpha signals. Nevertheless, the current study provides a solid foundational methodology for extracting actionable intelligence from the noisy, high-dimensional reality of modern electronic markets.

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A Appendix

A.1 Hyperparameter Configuration

To ensure the reproducibility of our experimental results, we detail the specific hyperparameter settings used for the models in this study. These parameters were determined based on standard defaults and preliminary validation checks.

Table 3: Hyperparameter specifications for XGBoost and the Hybrid Neural Network (MLP-Mixer).

Model	**Hyperparameter**	**Value**
XGBoost	Max Depth	6
	Learning Rate (η)	0.1
	Number of Estimators	60
	Early Stopping	None
Hybrid NN (GRU & Mixer)	Optimizer	AdamW
	Learning Rate	1×10^{-5}
	Batch Size	512
	Mixer Layers (Blocks)	2
	Sequence Length	60
	Hidden Dimension	64

The XGBoost model is trained using the `gpu_hist` tree method to leverage hardware acceleration. The Neural Network utilizes the AdamW optimizer, and the MLP-Mixer backbone is configured with a depth of 2 layers, which we found to effectively capture spatiotemporal dependencies without overfitting.